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Cyber Power as a Structural Variable of Maritime Power: A Normalized Multiplicative Model for Multidomain Assessment

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ABSTRACT The classical formulation of maritime power—the product of maritime interests and naval power—has become insufficient to explain the dynamics of digitalized maritime environments, where navigation, communications, traffic management, port facilities, and ocean surveillance depend on interconnected digital infrastructures. This article proposes an extended formulation that incorporates cyber power, strategic will, and maritime security as structural variables of maritime power. Following an analytical-conceptual approach, the model adopts a normalized multiplicative structure in which the collapse of any dimension constrains the aggregate, and whose output lies within the interval (0, 1], enabling a directly interpretable five-level scale. The formulation is verified through Monte Carlo simulation (100,000 trials per equation), confirming that every component equation preserves the required range. Colombia is used as an illustrative case study: anchored to the most systematic maritime power index currently available, the model reproduces the country's aggregate level and, on that basis, decomposes it to reveal that cyber power emerges as the most efficient strategic investment vector for middle regional powers, given its concave returns at low maturity levels. The article concludes by proposing a National Maritime Power Index as a research agenda, supported by an open-access interactive simulator that allows the model to be reproduced and explored.

INDEX TERMS Cyber defense, cyber power, cyber security, maritime domain awareness, maritime power, maritime security, strategic assessment, strategic will.

I. INTRODUCTION

THE maritime domain constitutes a decisive arena of contemporary strategic competition. For Colombia, a bioceanic state with 2,942 kilometers of coastline on the Caribbean Sea and the Pacific Ocean, approximately 988,000 km^2 of jurisdictional maritime territory, and eighteen recognized maritime interests, this dimension transcends the naval sphere and constitutes a factor of national development and security [31], [32]. The Naval Development Plan 2042 establishes as its second strategic objective "to protect and promote maritime and fluvial interests through naval power" [32, p. 58], synthesizing the logic upon which Colombian naval strategic thinking has been developed.

This conception has antecedents in the Western maritime strategic tradition. Since Mahan and Corbett, maritime power evolved toward a systemic understanding of the relationship among naval capabilities, maritime interests, and the strategic use of the sea. In the Latin American context, Solís Oyarzún [5] systematized this relationship, subsequently contextualized for Colombia by Uribe Cáceres et al. [7], who formalized maritime power as a function of maritime interests (MI) and naval power (NP). Within this evolution, Till [4] consolidated a systemic perspective in which both components mutually enable each other.

However, the digital transformation of the maritime domain introduces conditions that classical formulations did not anticipate. Navigation, communications, traffic management, port facilities, and ocean surveillance systems currently depend on interconnected digital infrastructures whose vulnerability can affect the exercise of maritime power [18], [23]. The 2017 cyberattack against Maersk, which disrupted its operations and generated estimated losses of USD 300 million [25], [26], and the cyber incident reported by the United States Coast Guard against a commercial vessel in 2019 [27], illustrate this transformation. In this context, Bebber [17] posits cyber power as a relevant component of contemporary

maritime power.

This transformation reveals a gap in the classical formulation: cyberspace does not appear as a structural domain of maritime power; strategic will is not formalized as a variable of the complete system; and Maritime Domain Awareness (MDA) is not differentiated from maritime consciousness as constructs of a distinct nature. Likewise, Vargas Suárez et al. [24], in one of the most systematic proposals for measuring maritime power, rank Colombia 33rd among 99 countries, while in maritime security it occupies the 75th position, evidencing the need to address additional dimensions of maritime power.

The research problem centers on the insufficiency of the classical formulation to represent maritime power in digitalized environments. The hypothesis holds that cyber power is a structural variable whose omission produces an incomplete representation of state capabilities and vulnerabilities. The objective is to develop an extended formulation that integrates cyber power, strategic will, and maritime security, with international theoretical scope and analytical illustration through the Colombian case. The model is accompanied by an open-access interactive simulator available at: https://github.com/diegocabuya/Ciberpoder-y-Poder-Maritimo.

The remainder of this article is organized as follows. Section II reviews related work on maritime power measurement and maritime cyber security, and identifies the research gap. Section IV presents the extended maritime power model and its constitutive equations. Section V reports the verification of the model's mathematical properties. Section VI applies the model to the Colombian case. Section VII discusses the implications and limitations, and Section VIII concludes.

II. RELATED WORK

A. MARITIME POWER AS A CONCEPTUAL SYSTEM

The modern conceptualization of maritime power finds one of its principal antecedents in the work of Alfred Thayer Mahan, who at the end of the nineteenth century analyzed the relationship among naval capabilities, maritime trade, strategic positions, and the ability of states to use the sea in support of their national interests [1], [8]. In *The Influence of Sea Power upon History, 1660–1783*, Mahan identified six principal conditions affecting a nation's maritime power: geographic position, physical configuration, territorial extent, population, character of the inhabitants, and character of the government [1]. Julian Corbett subsequently broadened this perspective by placing the control of the sea and of maritime communications, as well as the relationship between naval strategy and limited war, at the center of strategic analysis. Raoul Castex deepened this evolution by incorporating political, geographic, and material factors into a broader conception of maritime strategy [2], [3]. In the contemporary evolution of the concept, Geoffrey Till [4] frames maritime power as a system of interconnected elements, in which naval capabilities interact with the economic, political, and maritime components of the state, transcending a conception restricted to military power. This systemic perspective is particularly relevant for understanding the subsequent Latin American formulation of maritime power.

B. THE SOLÍS OYARZÚN FORMULATION AND ITS LATIN AMERICAN ADOPTION

In the Latin American strategic tradition, Eri Solís Oyarzún [5] systematized maritime power as a construct grounded in the interaction between maritime interests and naval power, understood as complementary components of a state's capacity to exploit, develop, and defend its interests in the maritime domain. This conception was subsequently made explicit in regional strategic literature: Schnaidt Mecklenburg [6] states that maritime power "is composed of Naval Power and Maritime Interests," a formulation underlying the expression $MP = MI \times NP$. Uribe Cáceres et al. [7] adopt and contextualize this structure for Colombia, additionally incorporating Maritime Consciousness—understood as the reflective knowledge of the sea and its possibilities—and Strategic Will—referring to the decision of state leaders in maritime and naval affairs—as dynamizing elements. González Núñez [9] documents the adoption of this conception in the Colombian strategic sphere and shows its subsequent evolution toward a broader interpretation of maritime power, while the Naval Development Plan 2042 incorporates this conception within the framework of national naval strategic planning [32].

C. MARITIME SECURITY IN THE CONTEMPORARY STRATEGIC ENVIRONMENT

Contemporary maritime security comprises a multidimensional spectrum of threats that transcends the traditional employment of naval power. Bueger and Edmunds [10] structure the maritime security agenda around three principal dimensions: interstate conflicts, including those associated with territorial disputes and gray-zone dynamics; maritime terrorism and other forms of violence; and blue crime, which comprises activities such as piracy, illicit trafficking, human trafficking, and environmental crimes, including illegal, unreported, and unregulated (IUU) fishing. In the international regulatory sphere, the International Maritime Organization (IMO) has established a comprehensive security framework whose central instruments are the International Ship and Port Facility Security Code (ISPS Code), in force since 2004, and the Convention for the Suppression of Unlawful Acts against the Safety of Maritime Navigation (SUA Convention) and its 2005 Protocols. The cyber dimension was subsequently incorporated into this framework through Resolution MSC.428(98), adopted in 2017, which mandated the integration of cyber risk management into shipping companies' safety management systems from 2021 onward.

The international architecture is complemented by regional cooperation mechanisms, among them the Yaoundé Code of Conduct in West and Central Africa and the Regional Cooperation Agreement on Combating Piracy and Armed Robbery against Ships in Asia (ReCAAP). In the Colombian case, Vargas-Suárez et al. [24] ranked the country 75th among

99 states in their maritime security index, with a value of 0.59, placing it within the group of countries with a lower relative level of maritime security. This result evidences the need to analyze maritime power not only from the availability of naval capabilities, but from its integral capacity to protect maritime interests against an increasingly broad spectrum of threats, including the cyber dimension.

D. CYBER POWER AS A STRATEGIC CATEGORY

Nye [13] defines cyber power as the ability to obtain preferred outcomes through the interconnected information resources of the cyber domain, distinguishing between cyber power as resources and as behavioral outcomes. Caro Bejarano [14] transfers the hard/soft power distinction to cyberspace: hard cyber power—coercive and denial capability—and soft—influence, norms, and cooperation. Their combination constitutes smart power—the effective articulation of hard and soft power—a paradigm relevant to middle powers with limited coercive capabilities, whose application to cyberspace, together with cyber diplomacy as an instrument of projection, is developed by Realpe Díaz [15]. Bebber [17] establishes that cyber power is a key element of maritime power: naval digital systems are vulnerable to attacks that degrade maritime power without physical confrontation. Espinel Bermúdez [18] documents the cyber dependence of the Colombian maritime domain, and Cabuya-Padilla [19], [20] develops the MARCIM framework.

E. STATE OF THE ART

The state of the art is organized around two research lines whose intersection defines the space of original contribution: (1) the formulation and measurement of maritime power, and (2) the incorporation of the cyber dimension into maritime power, from cyber security and cyber defense to the concept of cyber power.

1) Antecedents in the Formulation and Measurement of Maritime Power

The tradition of modeling maritime power has its most direct antecedent in Solís Oyarzún [5], subsequently systematized by Till [4] and adopted for the Colombian context by Uribe Cáceres et al. [7]. Uribe Cáceres [8] updates this formulation and presents maritime power (MP) as the result of the interaction between maritime interests (MI) and naval power (NP), with strategic will and maritime consciousness as dynamizing elements, without explicitly incorporating cyberspace as a structural component. This formulation constitutes the conceptual point of departure for examining the adequacy of the model in the face of the transformations of the digital strategic environment.

The most systematic attempt at quantitative operationalization corresponds to Vargas Suárez et al. [24], who develop a model applied to 99 countries. However, their proposal presents four gaps relevant to the purpose of this research: it does not incorporate cyberspace as a variable of maritime power or as a dimension of maritime security; it does not formalize strategic will as an independent variable of the system; it does not disaggregate maritime security by threat dimension; and it does not develop an explicit conceptual framework that supports the update of the index. These gaps limit the model's capacity to represent the new strategic conditions derived from the digitalization of the maritime domain.

2) Cyber Security, Cyber Defense, and Cyber Power in Maritime Power

The digitalization of the maritime domain has expanded the exposure surface of vessels, ports, navigation systems, communications, and critical infrastructure to cyber threats. Akpan et al. [43] report a 900% increase in cyber security breaches associated with operational technology (OT) systems, while Lagouvardou [22] identifies the convergence between information technology (IT) and OT as a relevant exposure factor. From a state-of-the-art perspective, Martínez et al. [21] document vulnerabilities associated with critical maritime systems, among them the Automatic Identification System (AIS) and the Electronic Chart Display and Information System (ECDIS). These works evidence that the digital transformation of the sector not only increases operational efficiency, but also the dependence on infrastructures whose disruption can compromise the continuity and security of maritime operations.

In response to these challenges, Cabuya-Padilla and Castaneda-Marroquin [19] present MARCIM—a Reference Framework for the Modeling and Simulation of Maritime Cyber Defense—aimed at conceptually structuring the maritime cyber defense domain and addressing gaps identified in the state of the art. Subsequently, Cabuya-Padilla et al. [20] develop SERDUX-MARCIM, a simulation model to analyze the propagation of cyberattacks in maritime infrastructure and to support cyber situational awareness at the strategic level. From a broader national security and defense perspective, Realpe and Cano [41] frame cyberspace as a strategic domain integrated with the traditional ones and propose approaching cyber defense from a systemic and multidimensional standpoint. Cano [39] deepens this perspective by linking defensive capabilities with the recognition of the adversary's offensive capabilities. Together, these works broaden the analysis from the identification of vulnerabilities toward the construction of frameworks and capabilities to understand and respond to cyber threats.

These works broaden the discussion from the protection of systems toward the strategic employment of cyber capabilities; along the same line, Realpe and Ayala [16] characterize cyber power as an articulation of technical, normative, defensive, and diplomatic capabilities aimed at consolidating digital sovereignty. Nevertheless, the reviewed literature does not identify a proposal that formalizes cyber power as a structural variable of a conceptual formulation of maritime power, nor one that establishes a normalized scale for its interpretation.

F. SYNTHESIS OF IDENTIFIED GAPS

Four convergent gaps delimit the contribution and scope of this study: (1) the absence of cyber power as a structural variable in the conceptual formulation of maritime power; (2) the absence of a formalization of strategic will as a variable of the maritime power system; (3) the aggregate treatment of maritime security, without disaggregation of its threat dimensions or explicit incorporation of cyberspace; and (4) the absence of a normalized conceptual framework that allows the formulation to be projected toward a National Maritime Power Index (NMPI) comparable across states.

III. METHOD

This article adopts an analytical-conceptual approach, understood as the systematic construction of an explanatory framework based on the critical synthesis of the existing literature, the identification of conceptual gaps, and the formulation of original theoretical propositions. This approach is consistent with the tradition of the field—Mahan, Corbett, Till, and Solís Oyarzún built their contributions through the systematization of strategic principles, without computing indices—which this article updates. The choice also responds to the fact that cyber power as a variable of maritime power is an emerging construct whose conceptual formalization must precede its quantitative operationalization.

The development followed four articulated stages. *First, a review of antecedents*: a search in Scopus, Web of Science, Google Scholar, and repositories of regional naval schools, with the terms "sea power," "poder marítimo," "cyber power," "maritime cybersecurity," "ciberdefensa marítima," "maritime domain awareness," and combinations thereof, prioritizing the period 2014–2024 without excluding seminal sources. *Second, a critical analysis of the classical formulation*: an in-depth study of Solís Oyarzún [5], Schnaidt Mecklenburg [6], Uribe Cáceres et al. [7], and González Núñez [9] to establish the content and analytical limits of $MP = MI \times NP$. *Third, the construction of the extended formula*: a two-level formulation with normalization through $n_{eff} = 3 + \gamma + \alpha$ [33], [34], justifying each functional form with the literature and anchoring the scale of the result to that of the Vargas Suárez et al. [24] index to guarantee its comparability. *Fourth, illustration through the Colombian case*: a qualitative-analytical analysis with empirical data from Vargas Suárez et al. [24] and Cabuya-Padilla [19], using Colombia as a reference case because it offers the most complete formulation and empirical data in the Spanish-language literature.

The model is conceptual and non-computational in nature: its parameters are calibratable and not universally fixed. Colombia is the illustrative case, not the sole scope of application. The specific limitations are detailed in the Discussion.

IV. THE EXTENDED MARITIME POWER MODEL

A. GENERAL EXPRESSION (LEVEL 1)

The classical formulation $MP = MI \times NP$ constitutes a multiplicative analytical expression that captures the structural interdependence between the interests a state holds over the sea and its naval capacity to defend them. Its conceptual strength resides in this nature: unlike an additive expression—in which a weak component can be compensated by a strong one—the product imposes that the collapse of either factor collapses the aggregate result. This property, the non-substitutability of structural components, is preserved and extended in this proposal. A critical analysis of the classical formulation in light of the contemporary environment reveals three structural variables whose omission produces an analytical gap with practical consequences.

Strategic Will (SW) operates, in the original formulation of Uribe Cáceres et al. [7], as an exclusive dynamizer of naval power, when in reality it conditions the complete system: political decision, institutional capacity, and the social valuation of the maritime domain equally determine the robustness of maritime interests and cyber maturity.

Maritime Security (MS) is treated by the classical formulation as a given context rather than as a structural variable of the system: a state with high MI and NP operating in an environment of severe maritime insecurity has an effective maritime power far below its potential.

Cyber Power (CP) becomes structural because the comprehensive digitalization of the maritime domain has transformed cyberspace into an enabling condition for the effective exercise of both naval power and maritime security, since the systems supporting both variables—C2 networks, AIS, ECDIS, GMDSS, surveillance networks—are cyber-dependent and therefore vulnerable to attacks that degrade maritime power without physical confrontation [17].

The proposed formula (Equation 1) incorporates the three omitted variables while keeping the classical core intact. All model variables take values in the interval $(0, 1]$, where 0 represents the total absence of the component and 1 represents the maximum theoretical level attainable. Here, SW denotes strategic will, MS the maritime security of the environment, and CP cyber power; the exponent γ (gamma) weights the global influence of maritime security, and α (alpha) that of cyber power:

$$MP = (MI \times NP) \times SW \times MS^{\gamma} \times CP^{\alpha} \tag{1}$$

Since Equation 1 employs a multiplicative aggregation of normalized components, the incorporation of multiple dimensions may produce aggregate values considerably lower than the individual component values. To obtain an aggregate index in the interval $(0, 1]$ and reduce the mechanical penalty associated with multiplying dimensions, a weighted geometric mean is adopted:

$$MP_{norm} = \left[(MI \times NP) \times SW \times MS^{\gamma} \times CP^{\alpha}\right]^{1/n_{eff}} \tag{2}$$

where the effective number of factors is:

$$n_{eff} = 3 + \gamma + \alpha \tag{3}$$

The normalization (Equations 2–3) is equivalent to computing the weighted geometric mean of the system, following the methodological principle applied by UNDP [33] to the Human Development Index and generalized by OECD [34] for composite indicators. It guarantees that a state with all components equal to 1 obtains $MP_{norm} = 1$ and keeps results within an interpretable range. The exponent n_{eff} equals the sum of all component exponents: $1\ (MI) + 1\ (NP) + 1\ (SW) + \gamma\ (MS) + \alpha\ (CP)$.

The architectural logic of Equation 1 responds to four explicit conceptual decisions. *(i) Preservation of the classical core*: the block $(MI \times NP)$ keeps the Solís Oyarzún [5] and Uribe Cáceres et al. [7], [8] formulation intact, recognizing its conceptual soundness and its documented institutional validity. *(ii) Strategic will as a system multiplier*: strategic will is not restricted to naval power—as in the original formulation—but operates as a multiplier of the complete maritime power system; with $SW \in (0, 1]$, $SW \to 0$ drives $MP_{norm} \to 0$, reflecting that without political, institutional, and social will, the potential generated by interests and capabilities does not materialize. *(iii) Maritime security as an environmental factor with exponent* γ: maritime security is incorporated as a condition of the strategic environment in which maritime power must materialize, with a calibratable exponent γ; following Bueger and Edmunds [10], a state with strong interests and capabilities but operating in a highly insecure environment sees its effective maritime power reduced. *(iv) Cyber power as a structural variable with exponent* α: cyber power is incorporated as the variable of greatest conceptual novelty, with an exponent α, the Strategic Cyber Maturity Index (SCMI), which, when it is < 1—the condition of most middle regional powers—makes the CP^{α} function concave.

In synthesis, the normalized formula expresses that the effective maritime power of a state in the twenty-first century is a function of the potential generated by its interests and naval capabilities, conditioned by political, institutional, and social will, and modulated by the security of the environment and by the state's capacity to operate and defend itself in maritime cyberspace.

B. INTERPRETATION SCALE

The normalization (Equation 2) produces results in $(0, 1]$ that allow a five-level analytical scale to be established for the strategic interpretation of maritime power. Table 1 presents this scale with its strategic description. The scale is constructed from the mathematical behavior of the normalized formula (Equations 2–3) and from the reference values verified for the generic validation profiles (Section V). The limits of each level are conventional and can be adjusted through empirical calibration.

The scale enables four direct applications: diagnosing the current strategic level of a state, identifying the target level of long-term planning, comparing middle powers within the same region, and exploring investment scenarios through the accompanying simulator.

TABLE 1. Interpretation Scale of Normalized Maritime Power

MP_{norm}	Level	Strategic description
0.81–1.00	High	Consolidated multidomain projection
0.61–0.80	Significant	Regional power, developed capabilities
0.41–0.60	Moderate	Developing power, identifiable gaps
0.21–0.40	Low	Incipient capability, structural vuln.
0.00–0.20	Critical	Structural maritime vulnerability

C. COMPONENT DISAGGREGATION (LEVEL 2)

Level 2 disaggregates each component into its constitutive variables, establishes their functional relationships, and justifies each modeling decision with explicit bibliographic support. All variables and sub-indices take values in $(0, 1]$.

1) Maritime Interests

Maritime interests constitute the component that defines the purpose of maritime power: what the state seeks to obtain, protect, or project through the ocean domain [28], [29]. Till [4] describes them as the set of motivations that lead states to concern themselves with the sea—resources, trade, communications, security, and the projection of influence—and establishes that their precise identification is a prior condition for any naval planning [7]. They are modeled as a weighted sub-index:

$$MI = \sum_{i=1}^{n} w_i \cdot MI_i \tag{4}$$

Equation 4 is a weighted sub-index where $MI_i \in (0, 1]$ represents the i-th maritime interest of the state on a normalized scale and w_i its relative weight according to the national strategic priority. The sum of the weights equals unity, guaranteeing $MI \in (0, 1]$. This formulation follows the composite-indicator methodology of the OECD [34], which recommends weighted aggregation when the dimensions have differentiable relative importance. For Colombia, $n = 18$ according to the Colombian Ocean Commission [28]; for other states, n varies according to the interests institutionally recognized. The weighting w_i is calibratable by state and period, which allows the sub-index to be updated as the strategic agenda changes without modifying the model architecture.

2) Naval Power

Naval power is the instrumental component of the system: the state's capacity to act on the maritime domain to defend its interests, deter threats, and project influence. Mahan [1] defined it as the synthesis of force and position, a formulation that Till [4] updates by incorporating technology and doctrine as transforming factors. Uribe Cáceres et al. [7] formalize the Colombian naval power in terms of platforms, armament, human resources, and deployment. The proposed equation introduces MDA as a third structural factor:

$$NP = F \times P \times MDA^{\mu} \tag{5}$$

F (force) integrates naval platforms, weapons systems, trained personnel, operational logistics, C2 systems, and tactical-operational maritime cyber defense capability—the aptitude of naval units to defend their own digital systems during concrete operations [7], [19]. P (position) captures the strategic geography, naval bases, area control, and access to maritime traffic routes. $MDA \in (0, 1]$ represents the operational awareness of the domain, with a calibration exponent μ. The decision to elevate MDA to the level of $F \times P$—instead of keeping it as a sub-component of F—responds to the argument of Bebber [17]: without situational awareness of the domain, force and position do not translate into effective naval power, because a state cannot defend what it does not perceive. MDA is distinguished from maritime consciousness (a component of SW): the former is a technical-operational capability, the latter a cultural-identity construct.

3) Strategic Will

Strategic will constitutes the component that determines the state's capacity to transform its maritime interests into sustained decisions, capabilities, and actions. In the formulation of Uribe Cáceres et al. [7], strategic will and maritime consciousness appear as dynamizing elements of naval power; this proposal reformulates strategic will as a multiplier of the complete system and operationalizes it into three dimensions through a weighted geometric mean:

$$SW = \left(PW^{w_1} \cdot IW^{w_2} \cdot SocW^{w_3}\right)^{1/\Sigma w} \tag{6}$$

PW (political will) represents the decision and orientation of state leaders to prioritize and sustain national maritime interests, expressed in the definition of strategic objectives, the allocation of resources, and the continuity of maritime policies. IW (institutional will) represents the capacity of the state's institutional apparatus to translate political decisions into stable norms, plans, and organizational structures. $SocW$ (social will) represents the capacity of society to transform maritime consciousness into appropriation, support, participation, and effective use of the maritime domain in the service of national interests. The proposed tripartition thus constitutes an original operationalization supported by the literature, through which strategic will integrates the capacity to decide politically, to institutionalize decisions, and to socially sustain them. The weighted geometric mean guarantees that the collapse of any dimension ($SocW \to 0$, for example) drives $SW \to 0$, reflecting that a will supported only politically, without institutional and social sustenance, is not strategically robust.

4) Maritime Security

Maritime security is the condition of the strategic environment in which the state must materialize its maritime power. Unlike the other components, MS is partially exogenous: it depends on the behavior of other state and non-state actors and on the dynamics of the regional environment. Following the agenda of Bueger and Edmunds [10], it is disaggregated into three threat dimensions through a weighted geometric mean:

$$MS = \left(IC^{a_1} \cdot TE^{a_2} \cdot BC^{a_3}\right)^{1/\Sigma a} \tag{7}$$

IC (interstate conflicts) comprises border, territorial, and maritime resource disputes among states, as well as gray-zone activities: coercive actions—the aggressive use of coast guards and fishing fleets—below the threshold of armed conflict. TE (terrorism and extremism) comprises attacks against vessels, platforms, and port facilities by extremist groups, as well as the use of the sea as a logistical corridor for terrorist financing. BC (blue crime) comprises the expressions of transnational organized crime at sea: crimes against maritime mobility—piracy and armed robbery—; transoceanic criminal flows—drug trafficking, human and arms trafficking—; and environmental crimes—IUU fishing and pollution. The exponents a_1, a_2, and a_3 (summing to unity) weight the dominant threat dimension according to the strategic profile of each state. The weighted geometric mean prevents a dimension of high security from compensating a critical dimension, and the global exponent γ in the general formula calibrates the weight of the environment over total maritime power.

5) Cyber Power

Cyber power is the component of greatest conceptual novelty and, for middle powers with $\alpha < 1$, the one with the highest initial returns per unit of investment at low levels of cyber maturity. Its incorporation as a structural variable is justified by the convergence of several independent lines: the strategic argument of Bebber [17], the empirical diagnosis of Espinel Bermúdez [18] and Martínez et al. [21], and the functional taxonomy of the MARCIM framework of Cabuya-Padilla [19], from which its three constitutive capabilities are drawn: maritime cyber security (CS_m), maritime cyber defense (CD_m), and maritime cyber resilience (CR_m). This disaggregation is consistent with the characterization of cyber power as an articulation of technical, normative, defensive, and diplomatic capabilities proposed by Realpe and Ayala [16], who conceive it as the active instrument through which a state consolidates its digital sovereignty—a strategic horizon that, in the maritime domain, implies the autonomous control of the infrastructure, data, and critical decisions of naval power. Cyber power aggregates its sub-components under a saturation operator:

$$CP = \min\left\{1,\ \left(\beta_1 CS_m + \beta_2 CD_m + \beta_3 CR_m\right)\left(1 + \delta \cdot CDip\right)\right\} \tag{8}$$

CS_m (maritime cyber security) is the protection of the digital assets that support the functioning of the maritime domain: AIS, ECDIS, GMDSS, naval C2 networks, digital port infrastructure, and surveillance platforms. CD_m (strategic maritime cyber defense) is the national capacity to operate and defend naval cyberspace at the level of policy, doctrine, and organization; it is supported by the state's cyberspace governance architecture—which articulates the Presidency, the defense

sector, and the joint cyber structures—and is grounded in the naval sphere; it is distinguished from tactical cyber defense (included in F within NP) by the level of analysis, which avoids duplication between NP and CP. CR_m (maritime cyber resilience) is the capacity to maintain the operational continuity of naval power and maritime domain services in the face of cyber incidents (detection, containment, recovery, and adaptation); it is the sub-component with the least institutional development in Colombia [30] and the one with the greatest marginal impact on national cyber power. *CDip* (cyber diplomacy) is the instrument that amplifies technical cyber power toward the international environment through diplomatic resources, including participation in the normative frameworks of the United Nations [42] and the Open-Ended Working Group (OEWG) [15]. The factor $(1 + \delta \cdot CDip)$ guarantees that cyber diplomacy never reduces cyber power—when $CDip = 0$, the factor is 1—and the saturation operator $\min\{1, \cdot\}$ in Equation 8 prevents the amplifying effect from exceeding the range, preserving $CP \in (0, 1]$. Cyber diplomacy also operates as an expression of political will in Equation 6, in a dual role without analytical duplication, with $\sum \beta_i = 1$.

The exponent α (SCMI) $\in (0, 1]$ calibrates with what intensity cyber power impacts MP_{norm} and absorbs the normative and governance frameworks, R&D investment, and organizational capacity that Cabuya-Padilla [19] identifies as determinants of maritime cyber defense. Its value admits a qualitative reading in five bands: incipient (0.00–0.20), with ad hoc capabilities and no formal frameworks; basic (0.21–0.40), with initial frameworks and fragmented governance; consolidating (0.41–0.60), with institutions and norms under development; advanced (0.61–0.80), with mature and integrated capabilities; and consolidated (0.81–1.00), characteristic of an international reference. When $\alpha < 1$—the condition of Colombia, in the consolidating band, and of most middle regional powers ($\alpha \approx 0.4$–0.7; ITU [35])—the function CP^{α} is concave: its marginal returns, high at low maturity levels, decrease as cyber power consolidates.

D. FULLY EXPANDED FORMULATION

Substituting Equations 4–8 into Equation 2 yields the fully expanded expression. All parameters are calibratable by state without modifying the fundamental architecture, and the output always satisfies $MP_{norm} \in (0, 1]$, interpretable through Table 1:

$$MP_{norm} = \left\{ \left[\left(\sum_{i=1}^{n} w_i MI_i \right) (FP\,MDA^{\mu}) \right] SW\,MS^{\gamma} CP^{\alpha} \right\}^{\frac{1}{3+\gamma+\alpha}} \quad (9)$$

V. MODEL VERIFICATION

To verify the logical-mathematical consistency of the model and explore its strategic properties, a three-stage validation was conducted: verification of logical-mathematical properties, sensitivity analysis, and verification through generic profiles.

For each component equation, 100,000 random trials were generated with inputs uniformly distributed over $(0, 1]$ and weights sampled from a Dirichlet distribution—thus satisfying the normalization constraints—to confirm that every equation preserves the required range. The verification confirms three properties. First, all component equations (4)–(8) produce values within $(0, 1]$ under the stated constraints. Second, the saturation operator $\min\{1, \cdot\}$ in the cyber power equation is necessary: without it, approximately 9.6% of the trials would exceed unity, confirming that the amplifying factor $(1+\delta \cdot CDip)$ can push the raw value beyond the range. Third, the expanded expression (Equation 9) is mathematically identical to the normalized form (Equation 2) when the Level-2 variables are substituted, with a maximum observed difference of 0.00×10^{0} across all trials.

The sensitivity analysis, using Colombia's reference values ($\alpha = 0.55$, $\gamma = 0.70$, hence $n_{eff} = 4.25$), specifies the strategic role of each variable: under equivalent percentage improvements (+20%), the direct-weight factors (*MI*, *NP*, and *SW*) exhibit the highest elasticity (+4.4% each), followed by *MS* (+3.0%) and *CP* (+2.4%), whose exponents dampen the proportional effect. The strategic priority of cyber power therefore does not derive from its point elasticity, but from three convergent properties: its lower baseline value (0.40, a greater margin for improvement), the concavity of CP^{α} with $\alpha < 1$ (high initial returns), and its transversal enabling character over naval power and maritime security, to which is added the asymmetry of cost and time relative to transforming maritime interests or naval power.

The verification through generic profiles produces results consistent with the scale of Table 1: the critical profile obtains $MP_{norm} \approx 0.19$ (Critical level), the reference profile $MP_{norm} \approx 0.52$ (Moderate level), and the consolidated profile $MP_{norm} \approx 0.85$ (High level), confirming that the model discriminates coherently across the strategic spectrum.

VI. CASE STUDY: COLOMBIA

Colombia constitutes the illustrative application case because relatively consolidated empirical information on its maritime power and maritime security is available. Vargas Suárez et al. [24] rank the country 33rd among 99 states in their maritime power index, with a value of 0.44, and 75th in maritime security, with a value of 0.59. Using these inputs as a reference and complementing them with the conceptual parameters of the proposed formulation, the model produces $MP_{norm} \approx 0.53$, which places the country at the *Moderate* level of the interpretation scale (Table 1). The consistency between this value and the reference index is not a validation in the strict sense—both models share the base data—but a calibration on a common scale that enables direct decomposition. On that basis, the model reveals a strategically relevant profile with two particularly relevant restrictions for Colombian maritime power: the security of the environment and cyber maturity.

Maritime security. The value of 0.59 reported by Vargas Suárez et al. [24] places Colombia in the 75th position among 99 states, in contrast to its 33rd place in global maritime

power. This gap evidences that the strategic environment systematically depresses effective maritime power relative to its potential. Applying Equation 7, the dominant dimension is blue crime, with drug trafficking through semi-submersibles in the Pacific, human trafficking in the Caribbean, and IUU fishing as the threats of greatest impact on the aggregate. Advancing from the Moderate to the Significant level therefore requires the elevation of maritime security as a necessary condition.

Cyber power. Cyber power introduces a second dimension of special interest. The reference value used for Colombia is 0.40, lower than that of the other principal components, which implies a relatively large margin for improvement. This component currently corresponds to the *consolidating* band ($\alpha \approx 0.4$–0.7; ITU [35]). The combination of a low baseline value and the concavity of the CP^{α} function makes cyber power the most efficient strategic investment vector for a middle regional power: at low maturity levels, each unit of investment produces higher marginal returns than in already-consolidated components, and its transversal character reinforces naval power and maritime security simultaneously.

Strategic will. Strategic will presents, finally, a qualitative dimension that complements the previous results. The Naval Development Plan 2042 [32] evidences advances in the definition of strategic objectives and in the institutionalization of the maritime interest, corresponding to a solid political and institutional will; the principal gap is located in social will—maritime consciousness translated into effective national appropriation—which the very literature identifies as a historical challenge for Colombia as a maritime nation.

Taken together, the Colombian application does not seek to externally validate the model, but to demonstrate its usefulness as a multidimensional diagnostic. The result allows the identification of the global level and the associated restrictions, as well as the prioritization of the areas of greatest strategic return.

VII. DISCUSSION

The results support the central hypothesis that cyber power can be conceptualized as a structural variable of maritime power in the twenty-first century, whose omission produces a systematic underestimation of vulnerabilities. On this basis, the contributions of the article can be ordered along three complementary planes.

On the *theoretical-conceptual* plane, the article formalizes cyber power as a fifth structural variable of maritime power and repositions strategic will as a multiplier of the complete system—and not only of naval power—while establishing the analytical distinction between maritime domain awareness (MDA), of a technical-operational nature, and maritime consciousness, of a cultural-identity nature. This core constitutes a reinterpretation of the classical theory from the perspective of a middle regional power with multidomain challenges.

On the *methodological* plane, the proposal contributes the normalization through the effective number of factors (n_{eff}), which guarantees that the resulting index remains within the interval $(0, 1]$, and a five-level interpretation scale that converts a numerical value into a comparable diagnosis. The formulation is accompanied by a verification instrument—the open-access interactive simulator—that allows the results to be reproduced and scenarios to be explored, in accordance with the principles of open science and replicability required of high-impact research.

On the *practical* plane, the model offers planners and decision-makers a tool to prioritize strategic investments in a grounded manner, identifying which dimensions concentrate the greatest gaps and which offer the most efficient returns. Its general character also allows the formulation to be projected toward a National Maritime Power Index (NMPI) that enables comparative studies among states with maritime projection.

The multiplicative structure, in contrast to an additive one, captures the non-substitutability of the components. Under a hypothetical collapse of maritime security, an additive model would reflect a reduction in aggregate maritime power of approximately 21%, whereas the multiplicative formulation reflects a reduction of approximately 33%, better representing the strategic reality that a severely insecure environment cannot be compensated by naval strength alone.

The model presents three explicitly declared limitations. First, its analytical—rather than fully calibrated—nature: the empirical estimation of weights through expert panels or the Analytic Hierarchy Process (AHP) constitutes the agenda of the NMPI. Second, the illustration is bounded to the Colombian case; extension to other regional navies would validate the robustness of the framework. Third, the model is static; a dynamic extension is feasible through simulation models such as MARCIM [20] to capture the temporal dimension of maritime power.

VIII. CONCLUSION

This article developed an analytical-conceptual update of the maritime power formula, extending the classical expression $MP = MI \times NP$—conceptually systematized by Solís Oyarzún [5] and subsequently formalized and adopted for the Colombian context by Uribe Cáceres et al. [7]—toward the normalized formulation MP_{norm}, interpretable within $(0, 1]$ through the five-level scale of Table 1. The proposal has international theoretical scope and analytical illustration through the Colombian case.

This dual scope constitutes the proposal's principal differential contribution. Unlike models formulated for a specific national context or that remain on a purely theoretical plane, the formulation developed here operates simultaneously on both levels: it is a general conceptual structure—applicable to any state with maritime projection, regardless of its condition as a global or middle power—and, at the same time, an instrument empirically anchored to a concrete case. The instantiation in the Colombian case does not exhaust the model nor particularize it; it functions as proof that the formulation is operationalizable with real data and produces interpretable results, which strengthens its transferability to other regional

scenarios, especially to middle maritime powers facing multidomain challenges with limited resources.

A second differential contribution resides in the nature of the conceptual leap with respect to the classical formulation. The work does not merely add a variable to a preexisting sum, but transforms the very logic of the construct: from an additive relation—in which a weak component can be compensated by a strong one—to a multiplicative architecture in which each dimension is structurally necessary and the collapse of any of them constrains aggregate maritime power. To this is added the transition from a non-operative theoretical statement to an index normalized in the interval $(0, 1]$, endowed with a five-level interpretation scale that translates an abstract value into an actionable strategic diagnosis.

Future work will focus on the operationalization of a National Maritime Power Index with empirically estimated weights, comparable across states with maritime projection, and on the dynamic extension of the model. The open-access interactive simulator that accompanies this article constitutes a first step toward the reproducibility and comparative exploration of the proposed framework.

CODE AVAILABILITY

The interactive simulator and all supporting materials are openly available at: https://github.com/diegocabuya/Ciberpoder-y-Poder-Maritimo.

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